MILITARY SPECIFICATION

BATTERY, PRIMARY RESERVE, AMMONIA, 5 MILLIAMPERE HOUR CAPACITY

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, manufacture and assembly of the battery used in the housing, timing and fuzing for the projectile, 155MM, M692/M731.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9298625 - Battery, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by Contracting Officer.

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and assemblies. - The parts and assemblies shall comply with all requirements specified on dwg. 9298625 and associated drawings and with all requirements specified in applicable specifications.

3.3 Ampule

3.3.1 Seal. - The ampule shall not show evidence of leakage when tested in accordance with 4.5.1.1.

3.3.2 Temperature cycling. - The ampule shall be capable of withstanding temperature cycling from minus sixty five degrees Fahrenheit plus or minus five degrees Fahrenheit to plus one hundred and seventy degrees Fahrenheit plus five degrees Fahrenheit ($-65^{\circ}F + 5^{\circ}F$ to $+170^{\circ}F +5^{\circ}F-0^{\circ}F$) when tested as specified in 4.5.1.2.

3.3.3 Ammonia composition. - The ammonia in the sealed and filled ampule shall not contain more than one tenth of one percent (0.1%) water.

3.3.4 Ammonia weight. - The weight of ammonia in a sealed ampule shall be sixty five plus thirty (65 + 30) milligrams.

3.4 Cathode.

3.4.1 Moisture content. - The cathode material shall not average more than one percent (1.0%) of water. No sample shall have a moisture content greater than one and five tenths percent (1.5%).

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3.4.2 Weight. - The cathode weight shall be seventy eight plus or minus six (78 ± 6.0) milligrams.

3.5 Battery primary reserve.

3.5.1 Case seal and pre-initiation voltage. - The battery shall be sealed to prevent leakage and/or permit an output voltage in excess of one-one hundredth (0.01) of a volt.

3.5.2 Activation. - The battery shall be initiated and activated in accordance with 4.5.9.2.

3.5.3 Rise time. - The battery output voltage shall rise to three and three tenths plus one and nine tenths (3.3 + 1.9) volts within ten (10) seconds when tested in accordance with 4.5.9.3. During the rise time, the output voltage shall not regress below two and two tenths (2.2) volts after attaining two and seven tenths (2.7) volts.

3.5.4 Short time capacity. - The battery shall be activated to the requirements of 3.5.2, meet the rise time requirements of 3.5.3 and have an output voltage of less than two and seventh tenths (2.7) volts after four (4) hours of continuous operation when tested as specified in 4.5.9.4.

3.5.5 Long time capacity. - The battery shall be activated to the requirements of 3.5.2, meet the rise time requirements of 3.5.3 and have an output voltage of three and three tenths plus one and nine tenths (3.3 + 1.9) volts for thirty eight (38) (Section I) to forty eight (48) (Section II) hours when tested as specified in 4.5.9.5.

3.5.6 Test temperature. The battery shall be capable of meeting all specified requirements at any temperature from minus thirty two degrees centigrade plus six degrees centrigrade to plus sixty degrees centigrade minus six degrees centigrade ($-32^{o}C + 6^{o}C$ to $+60^{o}C - 6^{o}C$).

3.6 Environmental and mechanical test

3.6.1 Thermal shock. - The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.1.

3.6.2 Temperature cycling. - The battery shall be capable of withstanding changes in temperature when tested in accordance with 4.5.10.2.

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3.6.3 Transportation vibration. - The battery shall be capable of withstanding mechanical shocks when tested in accordance with 4.5.10.3.

3.6.4 Drop test. - The battery shall be capable of withstanding shock when tested as specified in 4.5.10.4.

3.6.5 Mechanical shock. - The battery shall be capable of withstanding shock when tested as specified in 4.5.10.5.

3.7 Repair. - Any module failing to comply with all the test requirements and undergoes repair shall be tested as specified in 4.5.11.

3.8 Test temperature. - Unless specified otherwise, all testing shall be performed at a temperature of twenty five degrees centigrade plus five degrees centigrade ($25^{o}C \pm 5^{o}C$).

3.9 Data requirements. - Data shall be recorded and retained for all test required by this specification and as specified in 4.5.12.

3.10 Process controls. - All process controls essential for battery manufacuturer as deemed necessary by the Government to assure a reliable end item shall be submitted and maintained in accordance with 4.5.13.

3.11 Workmanship. - All mechanical assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agents. All markings and stampings shall be free of harmful cleaning and soldering agents that may be injurious to the assembly. All markings and stampings, either component or assembly shall be displayed and visibly defined. The battery shall be manufactured and/or processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

3.12 Qualification. - The battery shall be requalified if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the battery are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

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b. The contractor relocates his assembly facilities.

c. The contractor uses a new supplier for major sub-assemblies.

d. A change in material used for battery production or assembly.

Qualification requirements shall be determined by the Government for each change as specified in 4.4.3.14.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First article inspection.
b. Quality conformance inspections.

4.3 First article inspection

4.3.1 Submission. - The contract shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of:

PART DESCRIPTION	DRAWING NO.	QUANTITY
Ampule and Case assembly	9298626	25
Bulkhead	9298627	25
Ampule, Glass	9298628	75
Case	9298629	25
Separator	9298631-1	25
Separator	9298631-2	25
Cover and Collar Assembly	9298632	25

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Anode No. 1	9298633-1	25
Anode No. 2	9298633-2	25
Collar	9298634	25
Cover Assembly	9298636	25
Cathode	9298640	75
Cup and Anode Assembly	9298642	25
Cell Cup	9298645	25
Battery, Primary Reserve, Assembly	9298625	336
Support, Ampule	9298650	25

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ampule and Case Assembly (Dwg No. 9298626)				
	Presence of ammonia in ampule	25		3.2	4.5.8
	Bulk head to case force test	25		3.2	4.5.3
	Bulkhead (Dwg No. 9298627)				
	Examination for defects	25		3.2	4.4.2.2
	Ampule, Glass (Dwg No. 9298628)				
	Examination for defects	25		3.2	4.4.2.3
	Seal	25	3.3.1		4.5.1.1
	Temperature cycling	25	3.3.2		4.5.1.2
	Ammonia composition	25	3.3.3		4.5.1.3
	Weight	25	3.3.4		4.5.1.4
	Case (Dwg 9298629)				
	Examination for defects	25		3.2	4.4.2.4
	Separator (Dwg 9298631-1)				
	Salt content	25		3.2	4.5.7

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Separator (Dwg 9298631-2)				
	Salt content	25		3.2	4.5.7
	Cover and collar assembly (Dwg 9298632)				
	Weld strength	25		3.2	4.5.6.1
	Vacuum pull test	25		3.2	4.5.6.2
	Anode No. 1 (Dwg 9298633-1)				
	Examination for defects	25		3.2	4.5.2.6
	Anode No. 2 (Dwg 9298633-2)				
	Examination for defects	25		3.2	4.4.2.7
	Collar (Dwg 9298634)				
	Examination for defects	25		3.2	4.4.2.8
	Cover assembly (Dwg 9298636)				
	Examination for defects	25		3.2	4.4.2.9

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cathode (Dwg 9298640)				
	Moisture content	25		3.4.1	4.5.2.1
	Weight	25		3.4.2	4.5.2.2
	Composition	25		3.2	4.5.2.3
	Cup and anode assembly (Dwg 9298642)				
	Weld test	25		3.2	4.5.5.1
	Vaccum pull test	25		3.2	4.5.5.2
	Cell cup (Dwg 9298645)				
	Examination for defects	25		3.2	4.4.2.11
	Battery, Assembly (Dwg 9298525)				
	Examination for defects	25		3.2	4.4.2.12
	Case seal and pre-initiation voltage	25		3.5.1	4.5.9.1
	Thermal shock	3(a)		3.6.1	4.5.10.1
	Temperature cycling	3(a)		3.6.2	4.5.10.2
	Transportation vibration	3(a)		3.6.3	4.5.10.3
	Drop test	3(a)		3.6.4	4.5.10.4
	Mechanical shock	3(a)		3.6.5	4.5.10.5
	Activation	261 (b)	(c)	3.5.2	4.5.9.2

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rise time	261(b)(c)		3.5.3	4.5.9.3
	Long time capacity	261(b)(c)		3.5.5	4.5.9.5
	Short time capacity	75(d)		3.5.4	4.5.9.4
	NOTES: (a) The batteries shall be subjected to the five (5) environmental tests in the sequence shown. After environmental testing, the batteries shall be subjected to the electrical test denoted by (b) however the batteries will not require rotation during activation in the sequence shown. Twelve (12) each batteries shall be tested at the temperature listed in Table II of 4.4.3.7.4. If the test sample contains one (1) reject for activation, Rise time or Section I of the long time discharge requirements, a second sample of 36 batteries shall be tested. The second sample shall have passed the above criteria providing there are zero (0) rejects in this sample. The FAAT shall be rejected if the first sample or the combined samples contain two (2) or more rejects for activation, rise time or discharge requirements of 4.4.3.7.2.				
	(c) The batteries shall be subjected to the electrical test denoted by (c) and in the sequence shown. Seventy five (75) each batteries shall be tested at the temperatures listed in Table II of 4.4.3.7.4. Three (3) rejects shall be allowed for the activation and rise time requirements; three (3) rejects shall be allowed through the Section I discharge requirements of 4.4.3.7.2 and eight (8) rejects shall be allowed through the Section II discharge requirements of 4.4.3.7.3.				
	(d) Twenty five (25) each batteries shall be test at the temperatures listed in Table II of 4.4.3.7.4. One (1) reject shall be allowed for the requirements of 4.4.3.7.1 and one (1) reject shall be allowed for the requirements of 4.4.3.7.2. Three (3) rejects shall be allowed for the requirements of 4.4.3.7.3.				
	Support, Ampule (Dwg. 9298650)				
	Examination for defects	25		3.2	4.4.2.13
	Hardness test	Note e		3.2	4.5.4

NOTES:

e. A Coupon from the support ampule batch shall be used as the sample.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298626
4.4.2.1	Ampule and Case Assembly				NEXT HIGHER ASSEMBLY 9298647
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of ammonia in ampule		100%	3.2	4.5.8
102	Bulkhead to case force test			3.2	4.4.3.9/4.5.3
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298627
4.4.2.2	Bulkhead				NEXT HIGHER ASSEMBLY 9298626
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness from outside to bottom of spherical radius		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298628
4.4.2.3	Ampule, Glass				NEXT HIGHER ASSEMBLY 9298626
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Seal		0.40%	3.3.1	4.5.1.1
102	Temperature cycling		100%	3.3.2	4.5.1.2
103	Ammonia composition	5(a)		3.3.3	4.5.1.3
104	Depth from base to mid-point at top of shoulder		0.40%	3.2	Gage
105	Depth of center point		0.40%	3.2	Gage
106	Concentricity of tip		0.40%	3.2	Gage
107	Presence of ammonia in ampule		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:
(a) The lot shall be rejected if one (1) or more samples are classed defective.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298629
4.4.2.4	Case				NEXT HIGHER ASSEMBLY 9296626
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside radius at bottom		.40%	3.2	Gage
102	Thickness of bottom		.40%	3.2	Gage
Minor					
201	Flatness of flange		.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298632
4.4.2.5	Cover and Collar Assembly				NEXT HIGHER ASSEMBLY 9298647
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Vacuum pull test		100%	3.2	4.5.6.2
102	Weld strength			3.2	4.4.3.4/4.5.6.1
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Anode No. 1				9298633-1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Anode No. 2				9298633-2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Collar				9298634
					NEXT HIGHER ASSEMBLY
					9298632
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Web thickness		.40%	3.2	Gage
Minor					
201	Anode side inside diameter, minimum		.65%	3.2	Gage
202	Cover assembly inside diameter, minimum		.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Cover Assembly				9298636
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of projection diameter		0.40%	3.2	Gage
Minor					
201	Diameter of small outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298642
4.4.2.10	Cup and Anode Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Vacuum pull test		100%	3.2	4.5.5.2
102	Weld strength			3.2	4.4.3.3/4.5.5.1
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Cell cup				9298645
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total height		.40%	3.2	Gage
Minor					
201	Presence of slots and holes		.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48343 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Battery, Assembly				9298625
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of case		0.40%	3.2	Gage
102	Overall height		0.40%	3.2	Gage
103	Case seal (pre-initiation voltage)		100%	3.5.1	4.5.9.1
104	Activation			3.5.2	4.5.9.2
105	Rise time			3.5.3	4.5.9.3
106	Long Term capacity-Section I			3.5.5	4.5.9.5
107	Long Term capacity-Section II			3.5.5	4.5.9.5 / 4.4.3.7.2
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Support, Ampule				9298650
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness test			3.2	4.4.3.2/4.5.4
Minor					
201	Inside diameter, minimum		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Cathode (See dwg. 9298640)

4.4.3.1.1 Moisture content. - Major Defect. - Each batch of cathode material shall be sampled for the specified moisture requirements in accordance with 4.5.2.1. If the batch passes the test requirements, it may be stored for a maximum of ninety (90) days providing the maximum relative humidity at ambient does not exceed five percent (5.0%). The batch shall be rejected if the sample is classed defective.

4.4.3.1.2 Weight, Major defect. - Five (5) cathodes from each station shall be weighed each production shift in accordance with 4.5.2.2. The previously made cathodes shall be rejected if one (1) or more samples are classed defective.

4.4.3.1.3 Composition, Major defect. - Each batch shall be tested for the specified requirement in accordance with 4.5.2.3. If the batch passes the test requirements, it may be stored for a maximum of ninety (90) days providing the maximum relative humidity does not exceed five percent (5%) The batch shall be rejected if the sample is classed defective.

4.4.3.2 Support ampule hardness (See dwg. 9298650). - Major defect. - A coupon specimen from each support ampule batch shall be made and tested in accordance with 4.5.4. The lot shall be rejected if the coupon is rejected.

4.4.3.3 Cup and anode assembly weld strength. - (See dwg. 9298642) Major defect. - Each welder shall be monitored for control and deviation in accordance with the following procedures and as specified in 4.5.5.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after forty eight(48) consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

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The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in eighty (80) consecutive hours of production. If one (1), or more samples are classed defective, the prior two (2) hours production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.4 Cover and collar assembly weld strength, (See Dwg. 9298632) Major Defect. - Each welder shall be monitered for control and deviation in accordance with the following procedures and as specified in 4.5.6.1.

The last six (6) samples each hour of production shall be tested for the specified weld requirement. If one (1) or more samples are classed defective, prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production.

The sample size shall be reduced to three (3) samples per hour if no defects are found after forty eight (48) consecutive hours of operation. If one (1) or more samples are classed defective, the prior hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

The sample size may be reduced to two (2) samples every two (2) hours if there are no failures in eighty (80) consecutive hours of production. If one (1) or more samples are classed defective the prior two (2) hour's production shall be classed defective. The welding problem shall be corrected if a defect is found during production. If two (2) consecutive samples contain one (1) or more defects, then the sampling shall revert back to the six (6) sample size category.

4.4.3.5 Separator No. 1 salt content. - (See Dwg. 9298331-1), Major Defect. - Each batch of separators shall be tested for the specified requirements in accordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

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4.4.3.6 Separator No. 2 salt content. - (See Dwg. 9298331-2), Major Defect. - Each batch of separators shall be tested for the specified requirements in accordance with 4.5.7. Three (3) samples consisting of five (5) of each disc shall be tested. The batch shall be rejected if the sample is classed defective.

4.4.3.7 Battery, reserve, primary (See dwg. 9298647)

4.4.3.7.1 Activation and rise time. Major defect. - One hundred and twenty five (125) batteries shall be randomly selected from each lot containing less than ten thousand (10,000) units. The batteries shall be tested for activation in accordance with 4.5.9.2 and rise time in accordance with 4.5.9.3. The lot shall be accepted if one (1) or less defects are in the above sample.

If the test sample contain two (2) rejects a second sample of one hundred and twenty five (125) batteries shall be randomly selected from the lot. The units shall be tested as above for the activation and rise time requirements.

The lot shall be rejected if the first sample contains three (3) or more defects or if the combined total of both samples yields four (4) or more defects.

Test temperatures and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

One hundred and seventy five (175) batteries shall be randomly selected from each lot containing ten thousand (10,000) or more units. The samples shall be tested for activation in accordance with 4.5.9.2 and rise time in accordance with 4.5.9.3. The lot shall be accepted if one (1) or less defects are in the above sample. If the test sample contains two (2) rejects, a second sample of one hundred and seventy five (175) batteries shall be randomly selected from the lot. These units shall be tested as specified above for the activation and rise time requirements. The lot shall be rejected if the first sample contains three (3) or more defects or if the combined total of both samples yields five (5) or more defects.

Test temperatures and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.2 Long time capacity discharge section I. Major Defect. - For lots sizes less than ten thousand (10,000) units, the contractor shall submit a sample of one hundred and twenty five (125) batteries which have successfully passed the requirements of 4.4.3.7.1 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if one (1) or less samples are classed defective.

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If the test sample contains two (2) rejects, a second sample of one hundred and twenty five (125) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

The lot shall be rejected if the first sample contains three (3) or more rejects or if the combined total of both samples yields four (4) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.7.3.4.

For lot sized of ten thousand (10,000) or more batteries, the contractor shall submit a sample of one hundred and seventy five (175) batteries which have successfully passed the requirements of 4.4.3.7.1 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if one (1) or less samples are classed defective.

If the test sample contains two (2) rejects a second sample of one hundred and seventy five (175) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

The lot shall be rejected if the first sample contains three (3) or more rejects or if the combined total of both samples yields five (5) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.3 Long time capacity discharge, section II. - Major Defect. - For lot sizes less than ten thousand (10,000) units, the contractor shall submit a sample of one hundred and twenty five (125) batteries which have successfully passed the requirements of 4.4.3.7.2 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if four (4) or less samples are classed defective.

If the test sample contains five (5) or six (6) rejects, a second sample of one hundred and twenty five (125) batteries shall be randomly selected from the lot and tested as above for the specified requirements.

The lot shall be rejected if the first sample contains seven (7) or more rejects if the combined total of both samples yields ten (10) or more rejects.

Test temperature and the numbering samples per temperature and the numbering samples per temperature group shall be in accordance with 4.4.3.7.4.

For lot sizes of ten thousand (10,000) or more batteries, the contractor shall submit a sample of one hundred and seventy five (175) batteries which have successfully passed the requirements of 4.4.3.7.2 and test them in accordance with 4.5.9.5 for the specified discharge requirements. The lot shall be accepted if six (6) or less samples are classed defective.

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If the lot contains more than six (6) defects but less than thirteen (13) defects, a second sample of one hundred and seventy five (175) batteries shall be selected from the lot and tested as above for all specified requirements.

The lot shall be rejected if the first sample contains thirteen (13) or more rejects or if the combined total of both samples yields fourteen (14) or more defects.

Test temperature and the number of samples per temperature group shall be in accordance with the requirements of 4.4.3.7.4.

4.4.3.7.4 Test temperature and temperature groups. - The samples from paragraphs 4.4.3.7.1 through 4.4.3.7.3 shall be tested in accordance with Table II and Table III.

TABLE II

For sample size of 125 batteries

Test temperature	-32°C to -26°C	20°C to 30°C	54°C to 60°C
Samples/group	42	41	42

TABLE III

For sample size of 200 batteries

Test temperature	-32°C to -26°C	20°C to 30°C	54°C to 60°C
Samples/group	58	58	59

4.4.3.8 Ampule weight (See dwg. 9298628), Major Defect. - Twenty (20) samples shall be randomly selected from each lot and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be accepted if it contains no defects or rejected if it contains two (2) or more defects. In the event the lot contains one (1) defect, a second sample of twenty (20) ampules shall be randomly selected and tested for the specified requirements in accordance with 4.5.1.4. The lot shall be rejected if the second sample contains one (1) or more defects.

4.4.3.9 Bulkhead to case force test, (See dwg. 9298626) Major Defect. - The lot shall be tested 100% for the specified requirements in accordance with 4.5.3 when the contractor uses non-automated manufacturing techniques. Any assembly classed defective shall be removed from the lot.

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The sampling shall be reduced whenever the contractor uses a fully automated manufacturing technique that does not lend itself to process interruption. For this procedure, the contractor shall select the last six (6) assemblies produced during the previous hour. The samples shall be tested for the specified requirements in accordance with 4.5.3. The quantity produced during the hour represented by the sample size shall be classed defective if one (1) or more of the samples are classed defective.

4.4.3.10 Repair. - Major Defect - All batteries undergoing repair shall be processed in accordance with 4.5.11. Any battery classed defective shall not be returned to the lot.

4.4.3.11 Test temperature. - Major Defect - All batteries test shall be conducted at the specified test temperature. Any battery test performed otherwise shall be classed defective.

4.4.3.12 Data requirements. - Major Defect - All bettery test shall have the test data recorded in accordance with 4.5.12. Any material tested without the proper data acquisition shall be classed defective.

4.4.3.13 Process controls. - Major Defect - All process controls procedures shall be processed in accordance with 4.5.13. Any process control used that is classed defective shall be cause for lot rejection.

4.4.3.14 Qualification requirements. - The battery shall be inspected and tested for all the requirements specified in Table I unless specified otherwise by the technical agency listed in 6.5.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Ampule

4.5.1.1 Seal. - The ampule shall be isolated and completely submerged in a test solution of phenolphthalein for a minimum time of four (4) hours. The test solution shall be composed of twenty five cubic centimeters (25cc) of a master

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solution dissolved in one thousand (1000) cc of deionized water. The referenced master solution shall consist of one(1) gram of phenolphthalein dissolved in sixty (60) cc of ethyl alchohol diluted to one hundred (100) cc with deionized water.

The ampule shall be classed defective if the test solution surrounding the amplue changes to a red or pink color. (Non-destructive test)

4.5.1.2 Temperature cycling. - The ampule shall be temperature cycled as specified when mounted in an isolated fixture in the temperature chamber. The ampule shall be cycled five (5) times as follows:

a. Heat soaked for a minimum of one and five tenths (1.5) hours at a minimum temperature of plus one hundred and seventy degrees fahrenheit (+170°F).

b. The temperature shall be gradually reduced.

c. The ampule shall be cold soaked for a minimum of one (1) hour at a maximum temperature of minus sixty five degrees fahrenheit (-65°F).

d. This is the completion of one (1) cycle.

e. The ampule temperature shall be gradually raised and the start of the second cycle and each successive cycle shall commense. Each cycle shall be in accordance with steps a through c. The five cycles shall be completed within seventy two (72) hours.

Any ampule that cracks, ruptures, discolors or leaks shall be classed defective and removed from the lot. (Non-destructive test).

4.5.1.3 Ammonia composition. - The ampule shall be fractured in a closed container and the vapor shall be analyzed for the specified amount of water using a gas chromatographic method or equivalent. Any ampule failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.1.4 Ammonia weight. - The weight shall be determined by weighing the sample ampule and then fracturing it in a closed container. The vapor shall be allowed to vent and the

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residue weighted. The difference in weight between the original weight and the residue shall be considered the amount of ammonia contained in the submitted sample. Any ampule failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.2 Cathode

4.5.2.1 Moisture content. - The sample of cathode material shall be weighted and dried at room temperature for a period not less than two (2) hours in a vacuum at a pressure of not more than two hundred (200) microns. The sample shall be reweighed and the loss shall be calculated. Any sample failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.2.2 Weight. - The cathode shall be weighed in an analytical balance. Any cathode failing to comply with all of the specified requirements shall be classed defective (Non-destructive test).

4.5.2.3 Composition. - A sample shall be checked by an approved quantitative analysis method for the correct percentage of materials. Any sample failing to yield the proper percentage of material shall be classed defective (Destructive Test).

4.5.3 Bulkhead to case force test. - The assembly shall be examined for the specified requirement by monitoring the specified depth at the required force. Any assembly that fails to achieve either the force or depth requirements shall be classed defective (Non-destructive test).

4.5.4 Support ampule hardness test. - The coupon shall be tested for the specified requirement using a durometer. The coupon shall be classed defective if it fails to comply with all of the specified requirements (Destructive test).

4.5.5 Cup and anode assembly

4.5.5.1 Weld strength. - The cup and anode assembly shall be pulled apart to separation. If the parts separate without indicating a tear or a definite nugget the welded assembly shall be classed defective (Destructive test).

4.5.5.2 Vacuum pull test. - The assembly shall be secured and a vacuum probe shall be applied to the welded

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surfaces for a minimum period of five tenths (0.5) seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective (Non-destructive test).

4.5.6 Cover collar assembly

4.5.6.1 Weld strength. - The cover and collar assembly shall be pulled apart to separation. If the parts separate without indicating a tear or a definite nugget, the welded assembly shall be classed defective (Destructive test).

4.5.6.2 Vacuum pull test. - The assembly shall be secured and a vacuum probe shall be applied to the welded surfaces for a minimum period of five tenths (0.5) seconds minimum. If any plate separates during the vacuum pull, the assembly shall be classed defective (Non-destructive test).

4.5.7 Separator salt content. - The separator salt content shall be checked by quantitative analysis of the disc. The analysis shall include a positive test for a presense of potassium as a major cation. The disc shall be classed defective if it fails to comply with all of the specified requirements (Destructive test).

4.5.8 Presense of ammonia in ampule. - The ampule, prior to insertion into the case shall be visually monitored or sensed by an approved system of inspection equipment. Any ampule that is cracked, low in ammonia, contains discolored ammonia, or empty shall be classed defective (Non-destructive test).

4.5.9 Battery, primary, reserve

4.5.9.1 Case seal and pre-initiation voltage. - The sealed battery shall be placed in a solvent vapor at a pressure of one hundred (100) pounds per square inch for a minimum period of thirty(30) minutes. The vapor shall be made from the solvent of the batteries electrochemical system. Prior to eight (8) hours after the battery is removed from the vapor solvent the output voltage shall be measured. Any battery indicating an output voltage in excess of the specified maximum requirement when measured with a one megaohm input impedance digital voltmeter shall be classed defective and removed from the lot (Non-destructive test).

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4.5.9.2 Activation. - The battery shall be pre-conditioned at the applicable test temperature, placed in an approved test fixture and rotated about the Y-Y' axis (See Figure 1) at thirty six hundred plus or minus two hundred (3600 ± 200) revolutions per minute.

Surface "A" (See Figure 1) of the battery shall be deformed while rotating at the specified velocity using an approved activation system.

Deformation of surface "A" shall be within twenty-five hundredths (.025") inches of the X-X' reference. (See Figure 1). The battery shall be deformed forty to fifty-two hundredths inch (.040" to .052") by a spherical surface, or ball of a seven-thirty seconds plus or minus one-thirty second inch (7/32 ± 1/32) inch diameter.

The rotation shall be completely stopped at two plus or minus one (2.0 ± 1.0) seconds after deforming surface "A".

A battery that fails to activate on the spin fixture shall be removed and examined for a "No test" condition. A replacement battery may be activated immediately. Test results for the replacement battery shall be used for acceptance only when an analysis determines the failure to be related to the activating equipment. A "No test" battery must have its filled ampule intact and all its components present and properly assembled and cause of non-activation must be attributed to the approved test equipment.

Transfer time from the preconditioning temperature chamber to the test fixture shall be such that the test sample temperature shall not vary more than five degrees Centrigrade (5½C) Any battery failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.9.3 Rise time. - The battery output voltage shall be continuously monitored for the total test time. The test circuit shown in Figure 2 for the effective eight hundred (800) ohm load shall be used as the test load. The battery shall be classed defective if it fails to comply with all of the specified requirements. The sample shall be activated in accordance with the method specified in 4.5.9.2 (Destructive test).

4.5.9.4 Short time capacity. - The battery shall be activated in accordance with 4.5.9.2, and measured for its rise time in accordance with 4.5.9.3. The battery shall be monitored a minimum of once every fifteen (15) minutes for the output requirements on an approved data acquistition system with the test circuit shown in Figure 2 being used for the effective eight hundred (800) ohm test load for a minimum of

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four (4) hours. The battery shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test)

4.5.9.5 Long term capacity. - The battery shall be activated in accordance with 4.5.9.2 and measured for its rise time in accordance with 4.5.9.3 at the required test temperature.

The battery shall be loaded with the effective test load shown in figure 2 for four hundred and five seconds plus twenty (405 +20) seconds after activation at the required test temperature. The test load shall then be switched to the fifty thousand (50,000) ohm load shown in figure 2.

The output voltage shall be monitored a minimum of once an hour on a data acquisition system for a minimum of forty eight (48) hours.

The sample shall be classed defective if it fails to comply with all of the specified requirements. (Destructive test)

During first artible acceptance testing the contractor shall continuously monitor the battery voltage until it drops below 2.7 volts.

4.5.10 Environmental and mechanical tests

4.5.10.1 Thermal shock. - The battery, in an approved test vehicle, shall be tested in accordance with Test Method 113 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirements (Non-destructive test).

4.5.10.2 Temperature cycling. - The battery, in an approved test vehicle, shall be tested in accordance with Test Method 105 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirements. Note the humidity requirements are not required for testing.

4.5.10.3 Transportation vibration. - The battery shall be tested in accordance with Procedure I, Test Method 104 of MIL-STD-331. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirement (Non-destructive test). Test temperature groups shall be minus fifty two centigrade

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plus or minus two degrees centigrade ($-52^{o}C + 2^{o}C$), thirty degrees centigrade plus ten degrees centigrade ($30^{o}C + 10^{o}C$) and sixty degrees centigrade plus or minus two degrees centigrade ($+60^{o}C \pm 2^{o}C$).

4.5.10.4 Drop test. - The battery, in an approved test vehicle, shall be tested in accordance with Test Method 111 of MIL-STD-331 except the height of drop shall be seven (7) feet. The battery shall not activate during test but be capable of normal operation in accordance with 4.5.9.5. The battery shall be classed defective if it fails to comply with all of the specified requirements (Non-destructive test).

4.5.10.5 Mechanical shock. - The battery shall be mounted in an approved test vehicle and subjected to a seven (7X) gun firing utilizing an approved soft catch method. The battery shall not be activate during test (Non-destructive test).

4.5.11 Repair. - Any battery undergoing repair shall have the procedures submitted to the technical agency listed in 6.5 prior to performing the necessary work. Any battery repaired without prior Government approval shall be considered as non-conforming material and classed defective (Non-destructive procedure).

4.5.12 Data requirements. - The contractor shall submit a quality inspection test, demonstration and evaluation report detail in accordance with Data Item Description DI-R-1724 (tailored) giving all detailed inspection and test results for test performed as required by paragraph 4.4.3.7. Test report format shall be submitted to the technical agency listed in 6.5.

4.5.13 Process controls. - All essential process control procedures shall be documented and submitted to the technical agency listed in 6.5. Any battery component, sub-assembly or assembly undergoing a process control deemed necessary to assure a reliable end item by the Government that has not been previously submitted to the Government shall be classed as non-conforming material and classed defective. All changes to these process controls must be submitted to the Government prior to the incorporation of the process control.

4.5.14 Dimensional control of molded or plastic parts. - In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1) a sample of three (3) parts (as molded) from each cavity shall be fully inspected

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dimensionally to quality a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.1 through 4.4.2.14 after continuous production of each 5000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QA-T for approval prior to manufacture of equipment. The noted sub-paragrpahs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc mins) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensioanl stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QA-A-S. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.15 Dimensional control of stamped parts. - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to quality the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

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If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he present objective evidence to the Contracting Officer to substantiate the request.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level C. - The batteries shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing. -

5.2.1 Level C. - The batteries preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking shall comply with MIL-STD-1169.

5.4 Shipping. - When shipments from more than one lot are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The components and assembly covered by this specification are intended for use as the power supply for the housing, timer and fuzing for the projectile, 155MM, M692/M731.

6.2 Ordering data. - See Sections 6.1 and 6.2 of MIL-A-48078.

6.3 The Government as referrred to in this specification is the technical agency listed in 6.5.

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6.4 Submission of inspection equipment designs for approval. (See MIL-A-48078). - Submit equipment designs as required to:

Commander
ARRADCOM
Dover, New Jersey 07801
ATTN: DRDAR-QAA-I

6.5 Distribution of ammunition data cards, inspection and test results, process controls. - Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARRADCOM
Dover, New Jersey 07801
ATTN: DRDAR-QAA-Q

6.6 Government - Contractor drawing number

Cross Reference List

Government Drawing No.	Honeywell Drawing No.
9298626	28101676
9299628	28101087
9298628	28001654
9298629	28101085
9298631-1	28001648-001
9298631-2	28001648-002
9298632	28101677
9298633-1	28001646-001
9298633-2	28001646-002
9298634	28001090
9298636	28101086
9298640	28101679
9298642	28101678
9298645	28101636
9298625	28101674
9298650	28106651

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number; 1345-A053

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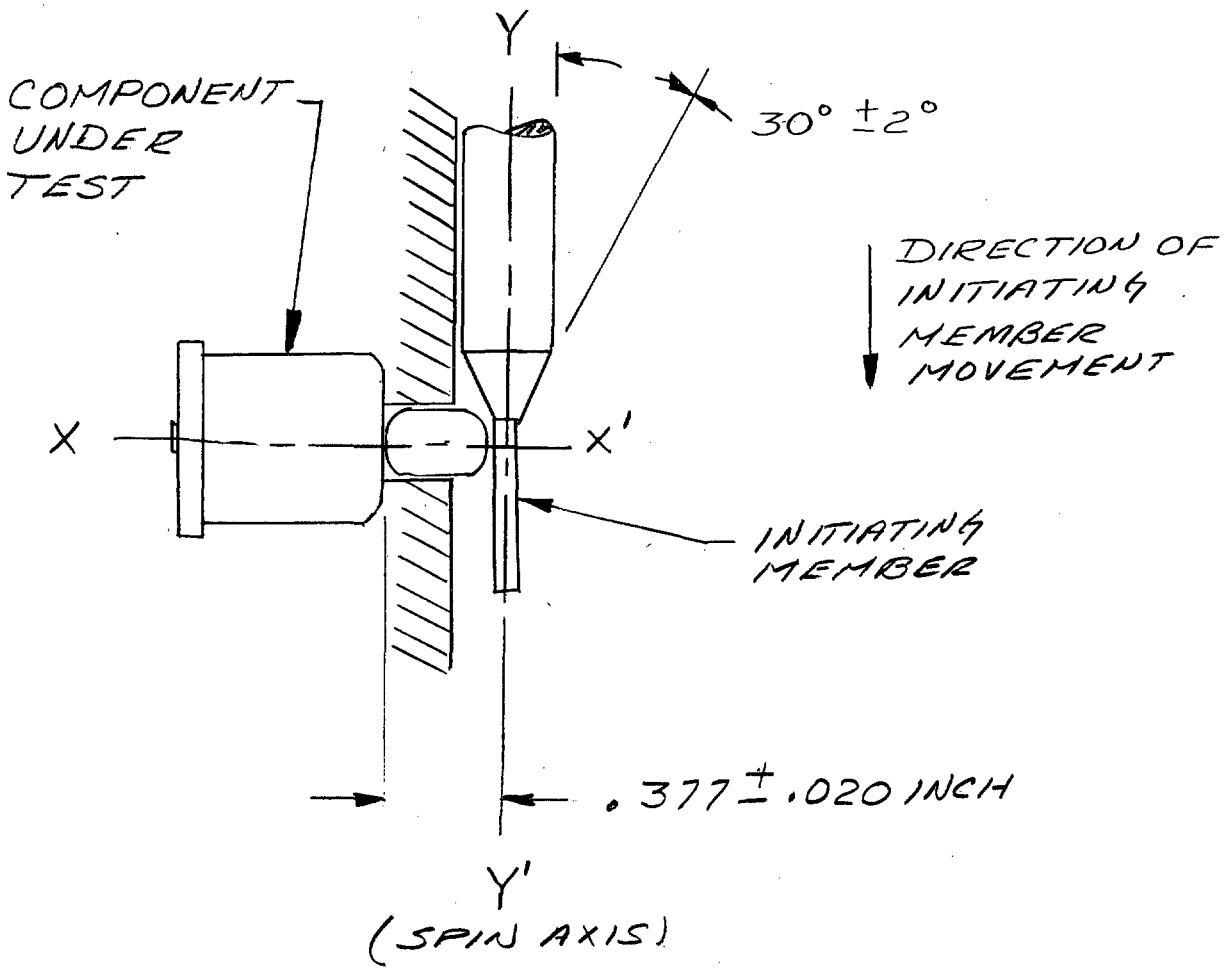


FIGURE 1

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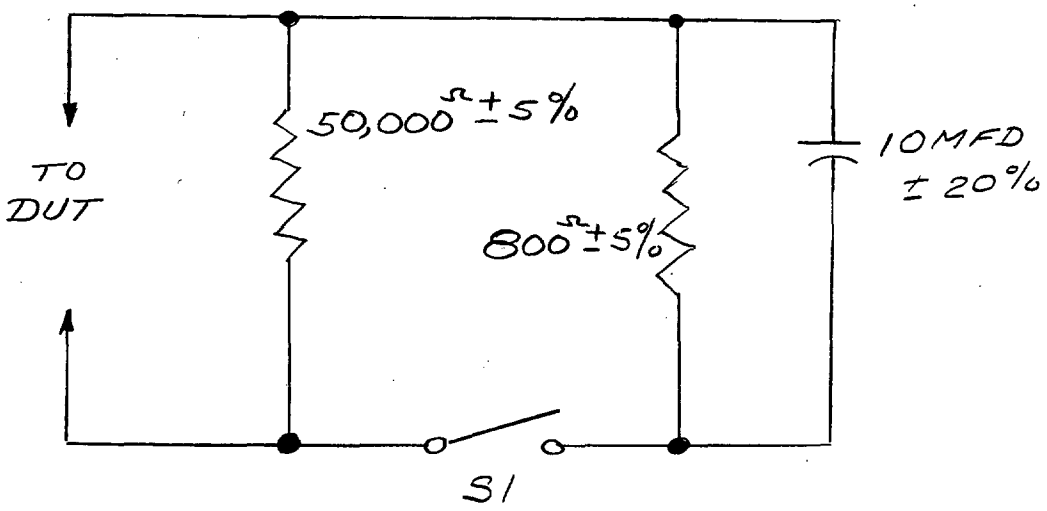


FIGURE 2

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INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-48343 Battery Primary Reserve Ammonia 5 Milliampere Hour Capacity

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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